cLACIr: Cognitive Load and Canine Intervention Recognition Dataset

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Abstract. Emotionally intelligent machines require the ability to differentiate emotional responses from humans from any background. Naturally, each human's emotional responses, whether it's physiological, facial, or any other modality, can be very unique to the individual. While these specific responses are unique though, there are underlying trends and patterns that can be understood when using larger sample sizes. The Cognitive Load and Canine Intervention Recognition (CLACIR) dataset consists of 140 subjects whose physiological responses to cognitive load and intervention are recorded using an Empatica E4. The dataset is split into two groups of subjects which are roughly equivalent in size and separated temporally. Physiological data consists of 3-axis accelerometry, electrodermal activity, and photoplethysmography recorded from the left wrist. This dataset is useful as a baseline for assessing machine learning models as well as for pretraining or mixing with other datasets.

Introduction

Human emotion and affect are physiological and behavioral reactions to changes in our environment [cite]. Some affective states may be easy to label (e.g., anger) but others may be more difficult to pin down (e.g., stress, anxiety). What's more, individuals vary greatly in how they experience affective states, with some experiencing intense emotions and other more muted emotions. The field of affective computing develops machines and algorithms that can detect human affective states using measures of physiological responses such as heart rate, breathing rate, skin conductance, and muscle activation [36]. These physiological data are input into algorithms that attempt to classify the affective state experienced when the data were collected. This has numerous

applications, including robots that are aware of human emotions when completing tasks [17], managing the workload of critical personnel [25], and improving treatment outcomes in therapy [56].

Due to a lack of larger datasets, many studies utilize the leave one subject out (LOSO) approach, which excludes one subject as the testing set and uses the remaining as the training set [44, 15]. This has preference because it maximizes the amount of data used in training while also implementing an identity split, where a separate person, whom the model has never seen, is tested. It is possible that the excluded subject is the subject that the trained models perform the best on, which would be an unrealistic estimation of both the amount of variance between subjects and the generalizability of selected features [30, 29]. One method to remedy this is to balance the representation of each subject and each class and to perform multiple splits of the dataset based on these representations, rather than just each class [29]. This method is called a stratified, group, k-fold split, which balances the class representation and the subject representation across k-folds [33].

Additionally, collecting physiological datasets can be difficult because subjects must either wear equipment during their daily life or participate for a short period using laboratory equipment [18, 47, 44, 19, 52, 25, 37, 14]. The time-series physiological signals being measured are generated by the human body through chemical and electrical interactions with various organs, which are largely involuntary and not easily controlled by an individual [19]. Consequently, targeted emotional states must either be generated in subjects within a laboratory setting or annotated through surveys from naturalistic settings by the subject. The labor required for these and other methods contributes to the low subject counts as well as to a low number of emotive states within the recorded data.

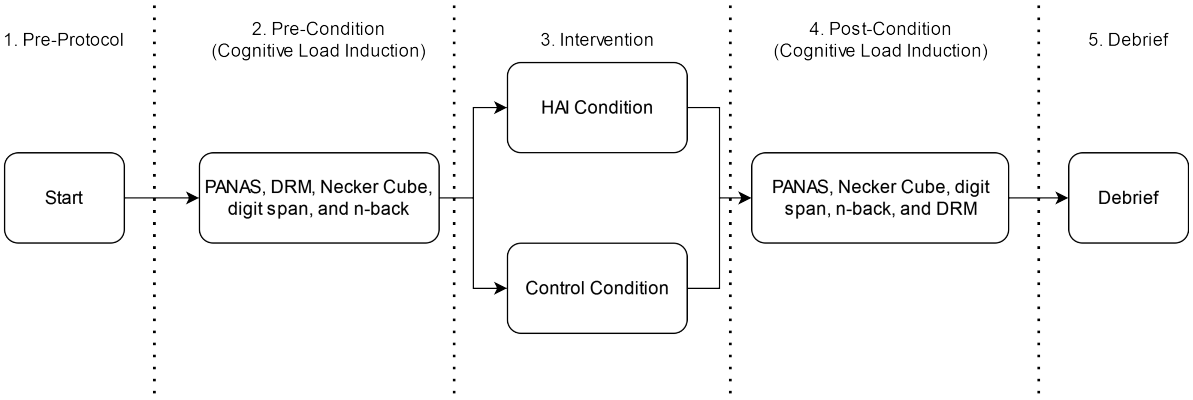

Figure 1. The experimental protocol to induce cognitive load states and provide intervention from the same using a control and human animal interaction (HAI) conditions.

The Cognitive Load and Canine Intervention Recognition (cLACIr) dataset consists of 140 subjects, all of whom are undergraduates in the University of Nebraska-Lincoln Department of Psychology subject pool [52]. In total, 95.8 hours of physiological data

are collected along with self-surveys of emotional state. Due to the size of this dataset, large identity splits can be achieved, providing a fair method of comparing different machine learning architectures. With cLACIr as a foundation, clinical applications can use machine learning for pilot studies in addition to traditional statistics [35], more robust machine learning models can be incorporated into digital domains like virtual reality [2], or can improve aggression detection approaches [19, 3].

Methods

Data Collection

This dataset was collected as a series of two experiments to investigate the effect of human animal interactions (HAIs) on affect and cognitive ability of the participants. The first experiment was conducted from September to November 2018 and recruited 73 subjects, 13 of which identified as male (17.8%) and 60 identified as female (82.2%), with a mean age of 19.2 (SD = 1.4) years old. The second experiment was conducted from November 2018 to April 2019 and recruited 83 participants, 17 of which identified as male (20.5%) and 66 identified as female (79.5%), with a mean age of 19.9 (SD = 1.8) years old. All participants were recruited from the University of Nebraska-Lincoln's psychology subject pool and did not have a physical or emotional aversion to dogs. Figure 2 is a schematic representation of the hierarchical structure of the dataset.

Before their arrival, each subject was pseudo randomly assigned to a human animal interaction (HAI) intervention condition or a control intervention condition, where each intervention condition is three minutes long. They are outfitted with an Empatica E4 [37] on their left wrist as they engage in cognitive tasks for long-term memory (Deese-Roedinger-McDermott task), attentional control (Necker cube pattern control task), working memory (backwards digit span task and n-back task). After these tasks are complete, they enter the intervention condition, after which they must complete the same cognitive tasks in a differing order.

To collect this data, an experimental protocol was designed that would induce a state of cognitive load on the participant, then apply an intervention condition, and then apply another cognitive load condition. The cognitive load induction has four separate cognitive tasks: Deese-Roedinger-McDermott (DRM) to test long term memory[41, 31], the Necker cube pattern control to test attentional control[32, 9, 42], the backwards digit span to test working memory[5], and the n-back to test working memory[40, 11]. These assessments were presented and responses collected using PsychoPy version 1.90.2[34] running on a computer's 16-inch monitor with only the participant and researcher present.

After the cognitive load state is induced in the participant, they move on to the intervention condition in a separate room. Each participant is pseudo randomly assigned to either a human animal interaction (HAI) condition or a control condition. For the HAI condition, the participant interacts with the second author's dog, who is a 65-

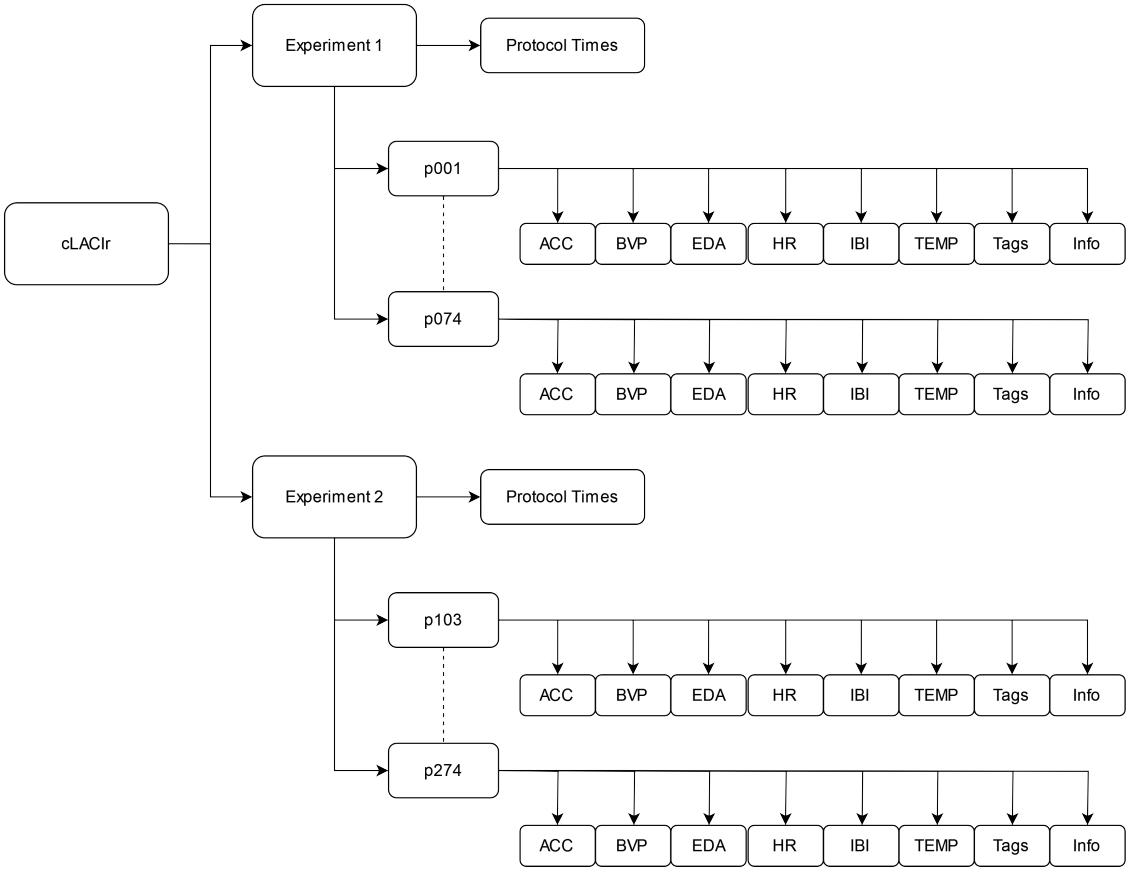

Figure 2. The hierarchical structure of the dataset.

pound, neutered, male Catahoula leopard mix that was Canine Good Citizen certified prior to the study. For the control condition, the participant is alone in the room and provided a sheet of paper with a full page of Latin text and instructed to circle every "e" and "f" for the "next few minutes". The participant was informed that the task would not be graded and there is no penalty for wrong answers. The intervention condition was performed for three minutes in both cases, based on the results from [21]. After the intervention condition, the participant is moved on to a second cognitive load induction activity.

To measure the participant's affect, the Positive and Negative Affect Schedule (PANAS)[54] was administered before each cognitive load induction. The second experiment also added additional measures of anxiety, including present and general feeling of anxiety with the State and Trait Anxiety Inventory (STAI) from [48] and their present feelings of anxiety using the Anxiety Visual Analogue Scale (AVAS) from [8]. Due to the new measures, the cognitive load induction tasks were modified to perform the AVAS first, then STAI, and then PANAS.

To measure the physiological responses of the participants as they go through the study protocol, which were not analyzed in previous studies due to low sampling rate of the heart rate signal [45, 52, 49], an Empatica E4 biosensor is affixed to their left wrist.

This wrist-worn biosensor has been used to generate data in many studies [19, 52, 44] and is regulatory compliant and validated [37], as well as being the top rated device for patient populations served by the behavioral analysis clinics [50]. This biosensor has four sensors: 64 Hz blood volume pulse (BVP) using photoplethysmography (PPG), 4 Hz electrodermal activity (EDA), 4 Hz infrared thermopile, and a 32 Hz 3-axis accelerometer measuring accelerations in its inertial frame of reference [19]. When transitioning between states of the protocol (start -> cognitive load induction -> intervention -> cognitive load induction -> debrief) the button on the biosensor is pressed, placing a marker in the datastream. This marker is used to label each data point with its associated protocol stage during post-processing.

Each physiological signal from the Empatica E4 encodes one of two response modalities from the subject. The first is the physical movement of the subject encoded by the accelerometer and the second is the autonomic nervous system responses to the environment. By looking at the movement of the subject's hand, specific movements that occur during emotional states can be assessed, which is the basis of physical activity recognition [39]. The separability of movements from different stimuli will likely be highly variable between subjects and may be highly correlated to the level of reaction [51].

Alternatively, the autonomic nervous system responses are highly correlated to internal emotional state [28] and is the basis of emotion recognition in machine learning systems [23]. The most common physiological signal used for this task is the EDA signal, which is collected by measuring the resistance between two contacts pressed against the subject's skin, which is the same principle as a lie detector [38, 2]. It can be decomposed into three constituent signals: the skin conductance response (SCR), the skin conductance level (SCL), and noise [20]. The noise will be ignored. The SCR is the fast-changing response to events, constituting a phasic component of the EDA signal, and can be used to approximate sympathetic nervous system activity [38, 4]. The SCL is a slower changing, tonic component of the EDA signal that acts as the baseline conductance of the skin. These two components indicate how strongly an emotion was felt and can also be used to differentiate between emotional states, for instance, the SCL and SCR will go up when someone is feeling anxiety and only the SCL will go up if someone is feeling happiness [28].

Peripheral skin temperature is a common measure of stress and can be used to distinguish between emotional states [12, 43, 26]. Generally, the theory of blood flow constriction due to stress causing a decrease in skin temperature and blood flow dilation due to happiness causing an increase in skin temperature is used [26]. But there are conflicting reports of skin temperature response to emotional states, where few report the temperature of the wrist or hand and instead focus on either monitoring facial temperature with a thermal camera [43] or will place their temperature probe in an area that is free from thermal environmental disturbances, such as under the armpit [26]. For studies that place the temperature probe on the hand [12], a decrease in skin temperature is observed during happy states and an increase in skin temperature is

observed for anger. Considering the placement of the Empatica E4, the observations from the temperature probe on the hand will be considered.

Finally, PPG is an indirect method of measuring the cardiac cycle and can be compared with the electrocardiogram (ECG), which directly measures the electrical activity of the heart [53]. PPG is performed by measuring the change in skin coloration as blood travels through the capillaries of the skin, which is directly correlated to the blood being pumped by the heart. From PPG, heart rate in beats per min (BPM) can be derived. In the case of someone feeling anxiety and happiness, their heart rate will go up [28].

The results of these methods and the statistical data analysis of the surveys are analyzed in [52] for the first experiment and in [49] for both. The physiological data has been analyzed and used for machine learning in the first author's master's thesis [1].

Data Preprocessing

To process the dataset, handcrafted features will be extracted from overlapping 60-second windows of data. These features consist of statistical information generated using the FLIRT toolbox [16]. The EDA signal is decomposed into its constituent signals using cvxEDA [20], where the noise is discarded and the SCL and SCR are processed using FLIRT, in addition to the original signal. The other signals (PPG, 3-axis accelerometry, and peripheral skin temperature) are processed in their original form using FLIRT. The calculated statistics for each signal window are the mean, standard deviation, minimum, maximum, dynamic range, sum, energy, skewness, kurtosis, peak count, RMS, line integral, number of samples above and below mean, number of sign changes, 25%-75% interquartile range, 5%-95% interquartile range, 5-th percentile, and 95-th percentile. Additionally, the entropy, perumutation entropy, and singular value decomposition entropy are calculated. All of these metrics are handled by FLIRT[16]. Each signal is broken into 5-second windows, which is a common approach [44, 35], and those windows have their statistical features calculated. Each participant's data will be tagged with their participant number, allowing the identity of the data to be tracked during training.

The labels are assigned based on the protocol steps for the first question, where the control intervention is considered neutral (label 0), cognitive load induction is considered stress (label 1), and HAI intervention is considered amusement (label 2), to align with [44]'s labeling. For the binary task, the cognitive load class and HAI intervention are merged. To assess how induced cognitive load can be reduced through different intervention conditions, a second question that will be posed is whether the machine learning models can differentiate between post-condition that are preceded by the HAI intervention or control intervention. The idea behind this question is that the level of cognitive load being experienced by the participant (and thereby recorded by the biosensor) is reduced in comparison to the control condition. Both of these questions will be repeated for the condition where the accelerometry features are removed. By

including these features, we hypothesize that it is likely to be a naive discriminator for the models, where HAI can be detected by wrist movements caused by petting the dog.

Machine Learning Approaches

The dataset will be benchmarked using a standard selection of common machine learning models. These models are Linear Discriminant Analysis (LDA), Decision Tree Classifier (DTC), Random Forest Classifier (RFC), and the K-Nearest Neighbors algorithm (KNP), all of which are implemented in SciKit-Learn [33]. An LDA model uses a linear combination of features to separate two or more classes, where predictions are made using Bayes' rule where the selected class maximizes the posterior probability [22]. These models are also closely related to analysis of variance (ANOVA), but the LDA uses continuous independent variables and categorical dependent variables [55]. The DTC model uses simple decision rules, such as feature a is less than feature b infers class a, which makes predictions very explainable [7]. The RFC is a meta-estimator that uses an ensemble of DTCs to predict classes based on the input data by taking a majority vote from the DTC ensemble [6]. Finally, the MLP is a neural network that multiplies the input data by an input weight matrix, which is passed through an activation function and then finally multiplied by an output weight matrix, which produces several outputs proportional to the data classes [24]. These models are trained by adjusting the weight matrices using various backpropagation algorithms, such as the Adam optimizer that moves the operating point of the model using first- and second-order moments [27].

To train the machine learning models, a stratified group 10-fold approach will be used [33]. Using the identity labels, dataset features, and dataset labels, 10 splits of the data will be generated, where each split contains a different group of identities and the number of samples from each class is as balanced as possible. By doing this, not only will the test set contain unique identities from the train set, a requirement for generalization performance, but the choice of identity for testing will also be evaluated. Doing this, a mean and standard deviation for each evaluation metric will be generated. If there are ideal and non-ideal choices for test participant, then this will be observed as a high standard deviation in the final metrics. Finally, the PANAS results are used to filter out subjects in the dataset by thresholding the difference between pre- and post-condition PANAS scores. By varying this from -1 to 1, we can filter out subjects that had an increasingly positive reaction to the data collection and see if the physiological data reflects this.

Results

HAI vs. Control Intervention Classification

The first classification test of the dataset was differentiating between the HAI condition and the control condition for the intervention stage of the data collection. The results

of this are shown in 3. The baseline value for balanced accuracy is shown by the black line, which is the ratio of one class over another.

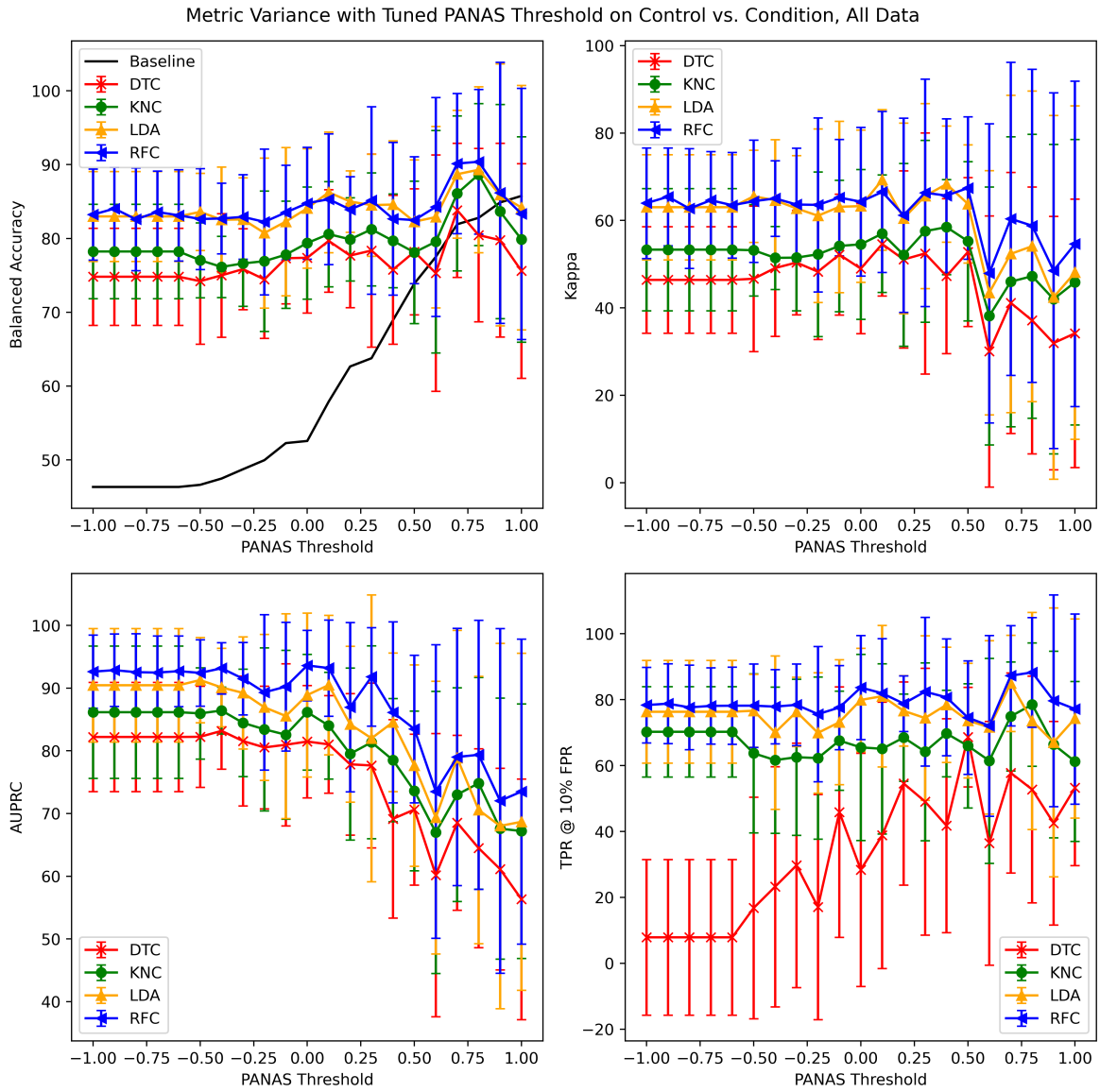

Figure 3. The effect of using various thresholds on the PANAS score to exclude subjects from dataset and training to differentiate between intervention conditions using data from intervention stage.

With the full dataset (PANAS threshold of -1), there are clear improvements that can be achieved by selecting different models. The RFC and LDA perform the best of the four, with balanced accuracy staying above 80% during the ablation study. There is even a slight bump to the balanced accuracy results when the PANAS threshold increases to around 0.5, indicating that removing subjects who had a more negative reaction to the study improves the ability to differentiate between classes. This is contrasted by the other metrics, where AUPRC continues to drop as the PANAS threshold increases.

The Kappa test shows some loss in agreement between raters, but the TPR @ 10% FPR shows a similar trend to the balanced accuracy, where it stays solid until the PANAS threshold is sufficiently high and it gets a boost. The models all show similar comparisons to one another across metrics (no model outperforms another in one metric that it underperforms in another), which is not entirely unexpected. Interestingly, the DTC model has a horrible TPR @ 10% FPR, which steadily increases until as the PANAS threshold is tuned.

HAI vs. Control Postcondition Classification

The second classification test of the dataset was differentiating between HAI and control conditions for the postcondition stage of the data collection. The results of this are shown in 4.

This classification task is much more difficult, because subjects are no longer in their intervention stage, so if they are in an excited state, then they are relaxing from it. This is reflected in the data, where the balanced accuracy for this task hovers slightly above the baseline performance. Even as the PANAS threshold is tuned, poor performance continues until around a PANAS threshold of 0.75, where a spike occurs. It doesn't reach the baseline value, but it increases to nearly 70% with all models. The same trends occurred as in 3 for the other metrics.

Discussion

Our dataset performs at the same level as other similar datasets. WESAD, with 15 participants, achieves upper 70% and lower 80% top-1 accuracy results across the board, the DT model performing the worst [44]. [18] had a wider variety of top-1 accuracy values, but achieved a peak value of 82.3% using their Snake dataset and 68.2% using their cognitive load dataset, each with 23 participants. These two examples shows that cLACIr either outperforms or is on par with other similar datasets. Through the sheer number of participants, our models show generalization in our task that is not able to be explored in other datasets due to limited participant count.

It is possible that there is a "cooldown" factor after a participant experiences the HAI condition, such that breaking up the postcondition classification by time bins would improve the accuracy. It is also possible that other machine learning models could perform better, for instance [18] uses XGBoost to achieve their top performing model on their Snake dataset. Additionally, there is the question of whether models trained with the cLACIr dataset would generalize to other datasets. Does the cognitive load task being learned by the best model also map to social evaluative stress from [44] or the cognitive load in [18]?

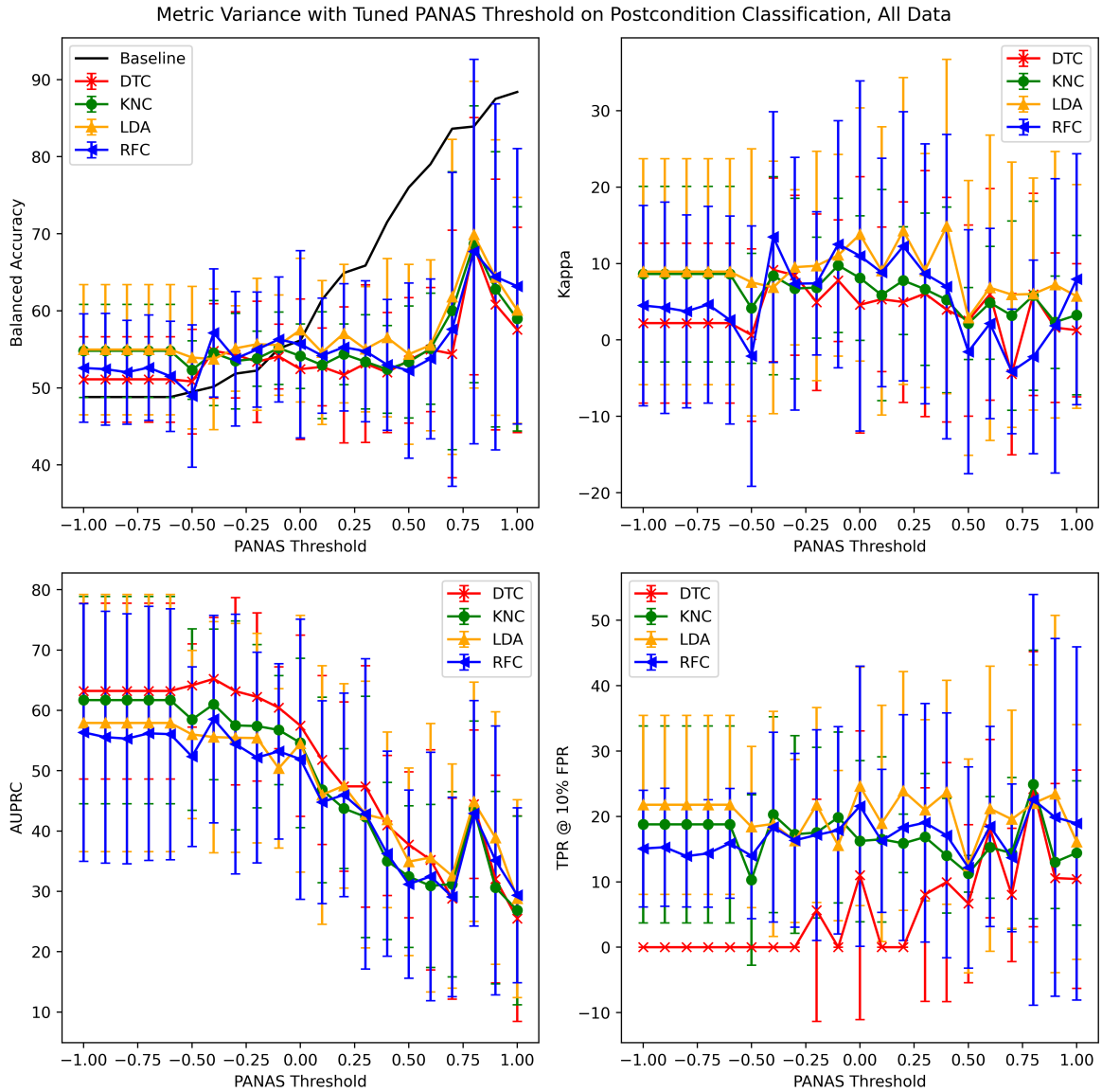

Figure 4. The effect of using various thresholds on the PANAS score to exclude subjects from dataset and training to differentiate between intervention conditions using data from postcondition stage.

Conclusions

This dataset provides a large sampling of subjects with a rigid and well-defined protocol that induces a commonly investigated mental state of cognitive load. This contrasts with common physiological datasets, where the main goal is to induce heightened states of emotion through Trier Stress Tests [44] or using provocative video [46]. The other issue is that most datasets are of a small sampling of people, usually less than 40 individuals and more commonly around 15. This leads to methodological issues where the generalization performance of the models cannot be assessed in a fair way. The

most common way of assessing generalization performance is the Leave One Subject Out approach, which can introduce bias by selecting the best performing subject to report. By having a very large dataset like cLACIr with 140 individuals, even a 10% train-test split leaves 14 individuals in the test set, which is nearly the same size as WESAD [44].

In addition, this dataset performs well on a variety of machine learning models, showing that the class labels are aligning well with the physiological data features and that learning is occurring. This is further enhanced by using multiple rich metrics that give insight into other aspects of the model's performance.

Ethical Statement

All data was collected and processed in an ethical manner, in full compliance with the UNL Internal Review Board's (protocol #19552) and Institutional Animal Care and Use Committee's (protocol #1599) relevant codes of experimentation and legislation. All participants gave written consent to participate and acknowledged that de-identified data could be published publicly.

Acknowledgments

Author contributions statement

Competing interests

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